

13. Which of these is not a product of fermentation?
 A) Lactate B) Oxygen
 C) Carbon dioxide D) Ethanol
14. Glucose molecule during the process of glycolysis is broken down into
 A) Four pyruvic acid B) Three pyruvic acid
 C) Two pyruvic acid D) One pyruvic acid
15. Which of these is a secondary metabolite?
 A) Proteins B) Lipids
 C) Vitamins D) Steroids
16. Which of these secondary metabolites is useful for human welfare?
 A) Curcumin B) Abrin
 C) Ricin D) DNA
17. A major ingredient of penicillin production media is
 A) Beef extract B) Yeast extract
 C) Corn steep liquor D) Protein meal
18. Which of the following is not the biofertilisers producing bacteria?
 A) Rhizobium spp B) Anabaena
 C) Clostridium D) Nostoc
19. The population of algae in soil is _____ that of either bacteria or fungi.
 A) generally smaller than B) generally greater than
 C) equal to D) none of these
20. An example of a symbiotic nitrogen fixer and is
 A) *Rhizobium* spp B) *Beijerinckia* spp
 C) *Clostridium* spp D) *Azotobacter* spp

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With diagram explain the structural and functional entities of the bacterial cell.	8	L*2	1	1,2
	b) Enlist Robert Koch's Postulates and his important contributions.	8	L1	2	1,2
2.	a) Define disinfection. Classify various chemical methods of sterilization with suitable examples.	8	L2	2	1,2
	b) Give an explanatory note on Viral replication.	8	L2	1	1,2
3.	a) What is molecular characterization? Describe any two major methods which can be used under this approach.	8	L1	1	1,2
	b) With diagram explain the working principles of Transmission Electron microscope (TEM).	8	L2	2	1,2
Unit – II					
4.	a) Discuss in detail the pathway converting glucose to pyruvic acid.	8	L2	4	1,2,7
	b) What is generation time? Summarize different phases of Growth curve.	8	L2	3	1,7
5.	a) Classify culture media based on application or chemical composition.	8	L1	3	1,7.
	b) Comment on anaerobic respiration in microorganisms.	8	L2	4	1,2,7
6.	a) Enlist and write about different types of methods used to measure bacterial growth.	8	L2	3	1,7
	b) Differentiate between primary and secondary metabolites of microorganisms and add a note on its significance.	8	L1	4	1,2,7

Unit – III

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|-------|---|---|----|---|-------|
| 7. a) | Comment on advantages of Liquid Bio-fertilizer over conventional carrier-based Bio-fertilizers. | 8 | L1 | 5 | 1,2,6 |
| b) | With flow chart explain about upstream and downstream processing to obtain penicillin antibiotic. | 8 | L2 | 5 | 1,2,6 |
| 8. a) | What are the common methods used for sewage treatment?
Elaborate | 8 | | | 1,2,6 |
| b) | With illustration explain the production of Cheese | 8 | L2 | 5 | |
| | | | L1 | 5 | 1,2,6 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
